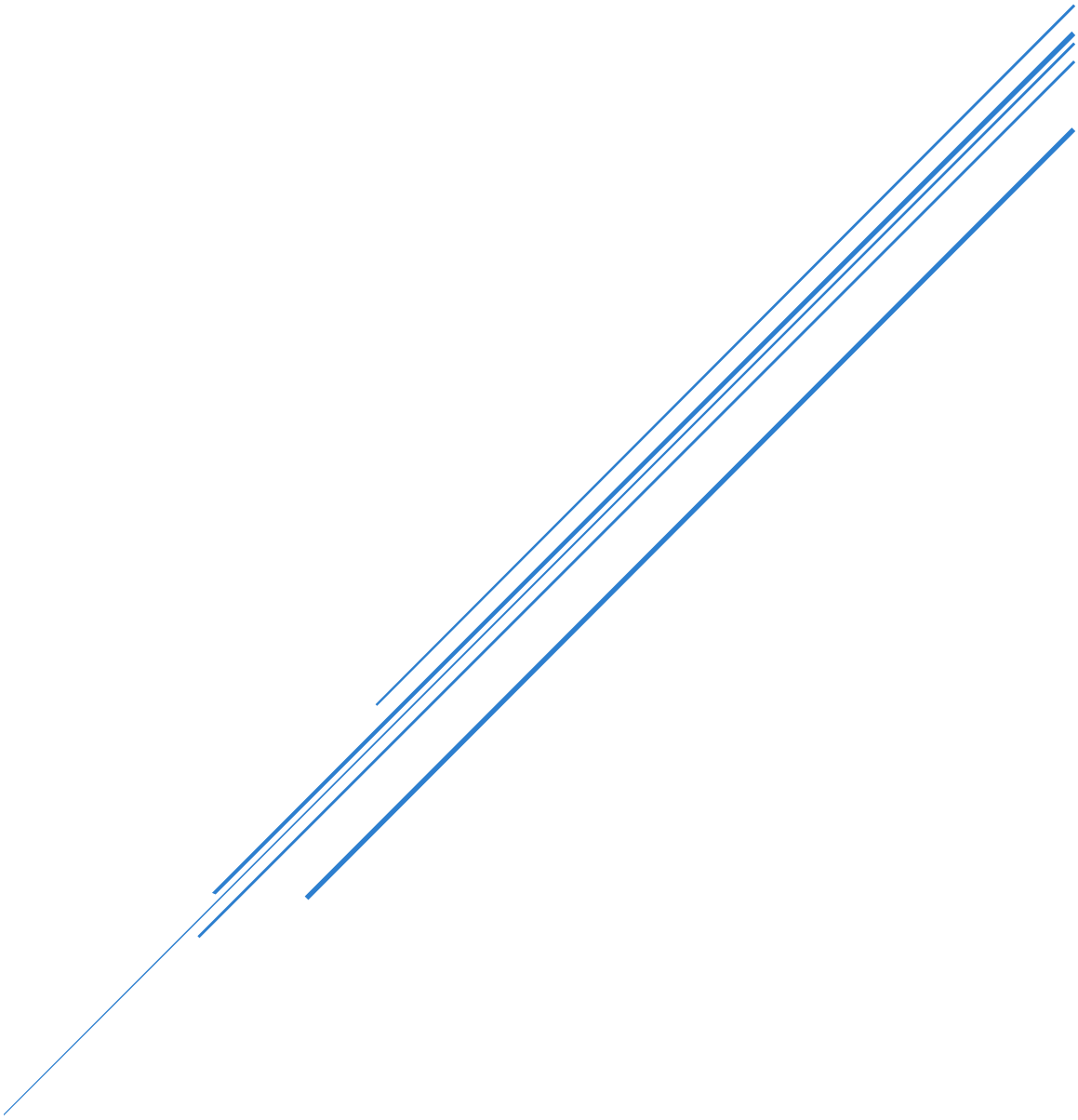
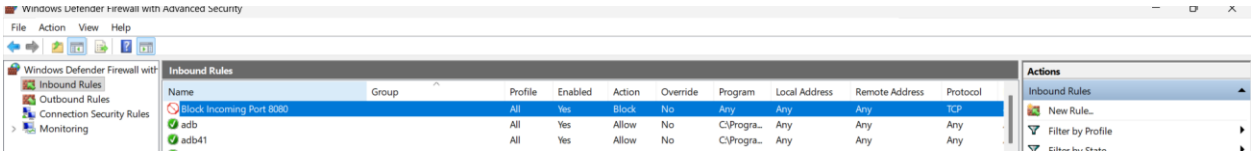
SECURE NETWORKS WITH FIREWALLS NAT, POT FILTERING, PACKET FILTERING, AND OTHER METHODS

Assignment - 2



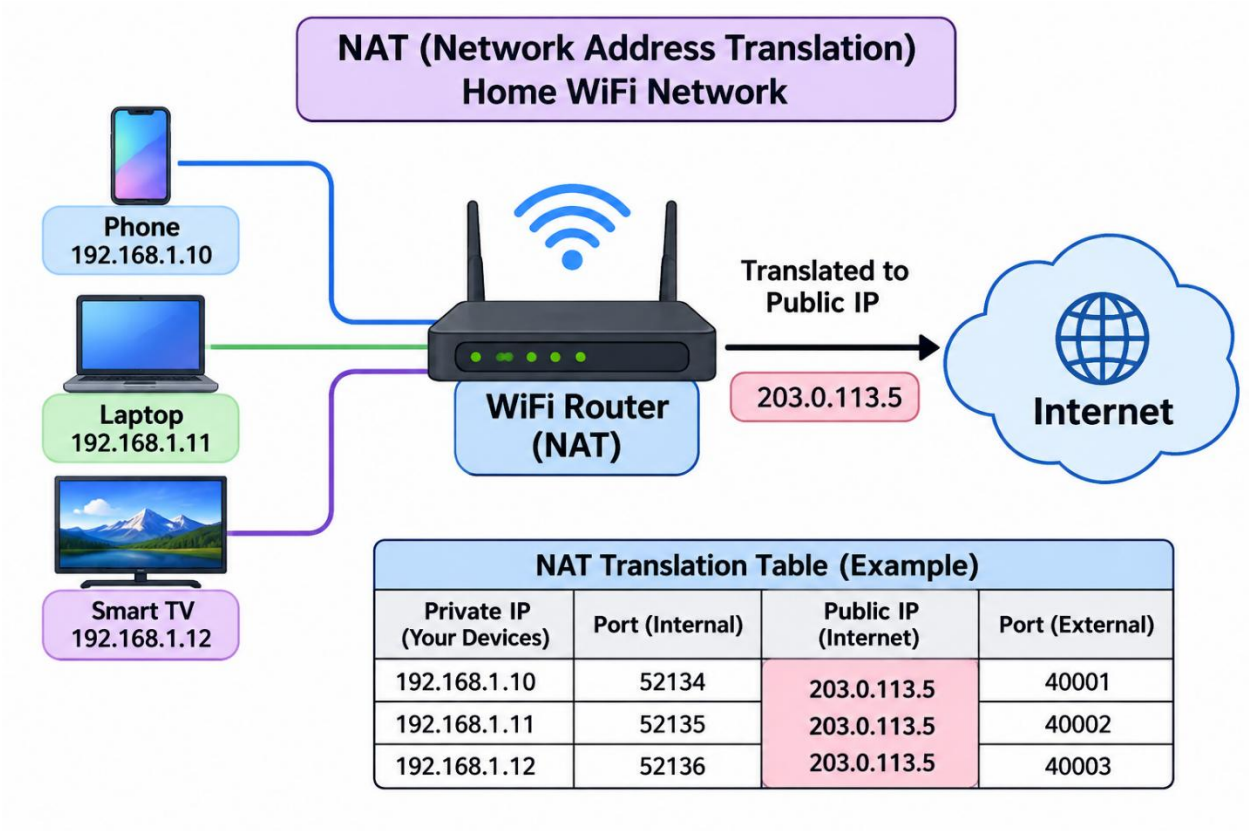
1. **Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.**

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2. **Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.**
   Hint: Use any drawing tool or even hand-draw and click a photo.

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NAT Translation Table (Example)

Private IP (Your Devices)	Port (Internal)	Public IP (Internet)	Port (External)
192.168.1.10	52134	203.0.113.5	40001
192.168.1.11	52135	203.0.113.5	40002
192.168.1.12	52136	203.0.113.5	40003

NAT is a process used by a WiFi router to translate private IP addresses into a single public IP address. The router allows multiple devices, such as a phone, laptop, and smart TV, to access the internet using one public IP address.

Example:

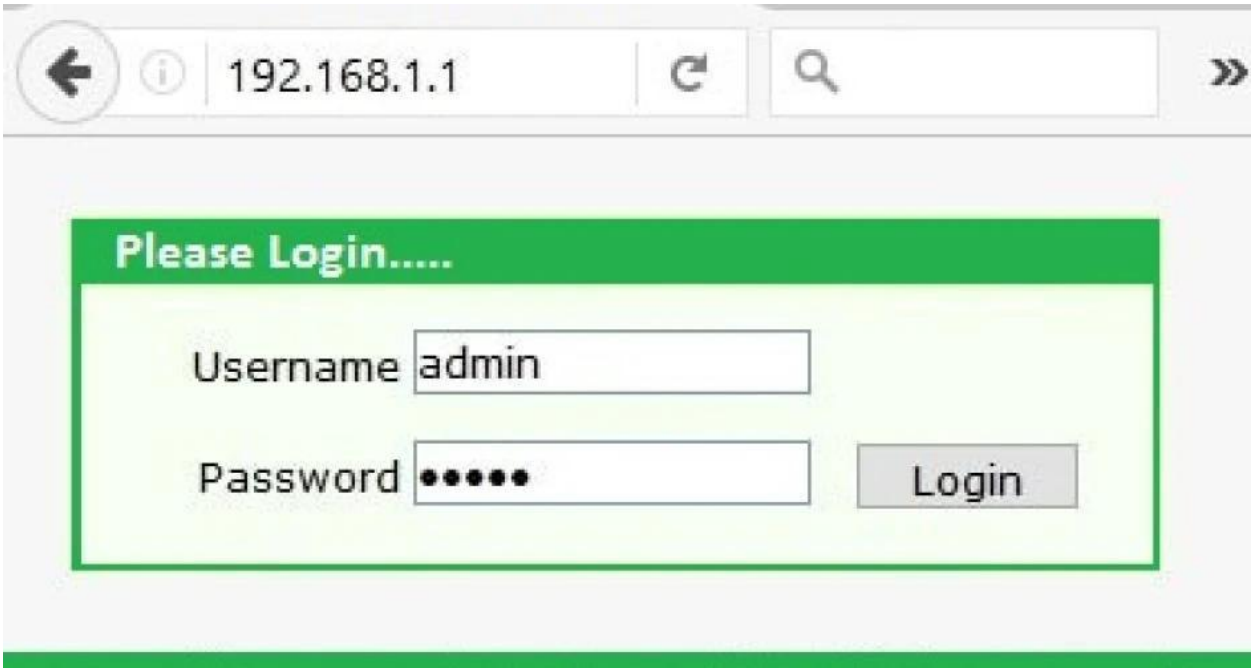
- **Phone: 192.168.1.10**
- **Laptop: 192.168.1.11**
- **Smart TV: 192.168.1.12**
- **Router Public IP: 203.0.113.5**

3. **Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.**

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Step 1: Open Router Admin Panel

1. **Connect your laptop to your WiFi.**
2. **Open Chrome.**
3. **Enter your router's gateway IP address:**
    - **192.168.1.1 or 192.168.0.1**
4. **Enter your router's admin username and password.**

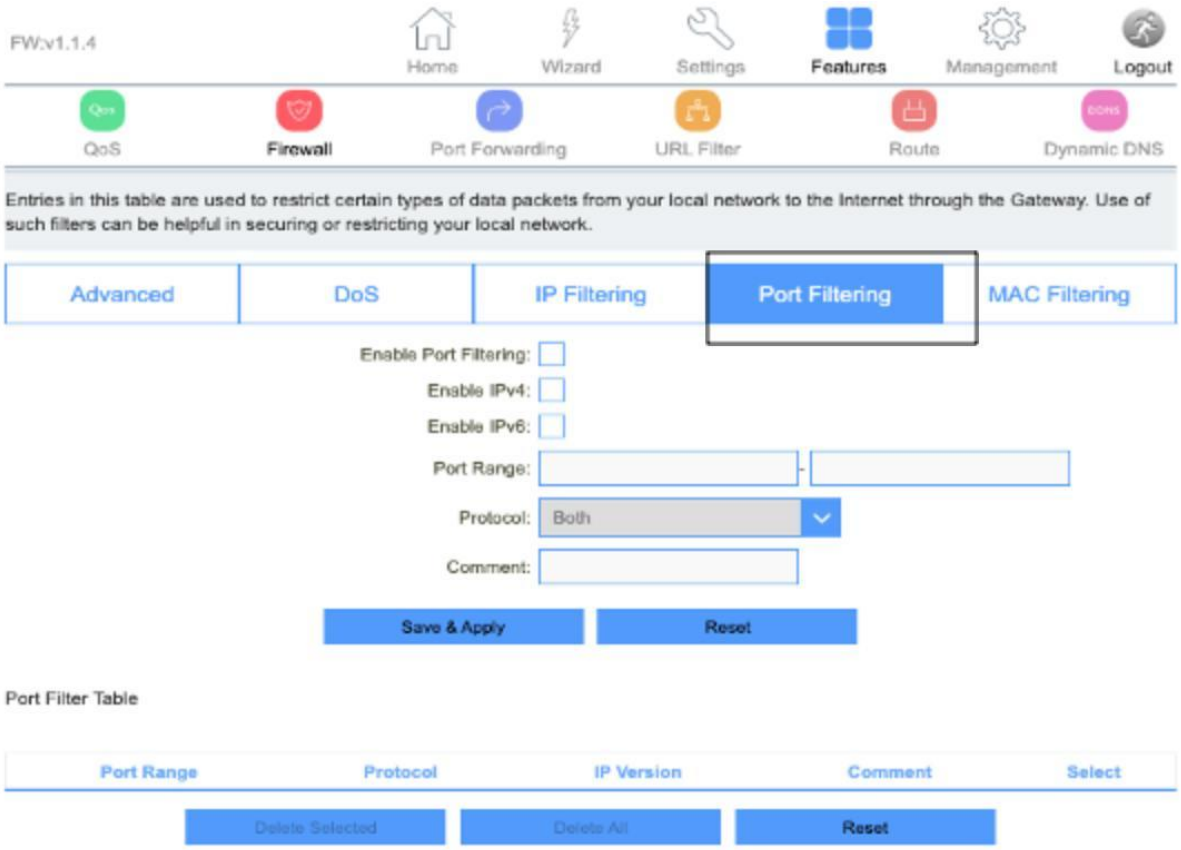


Step 2: Find Port Filtering

Look for one of these options in the router settings:

- **Security**
- **Firewall**
- **Access Control**
- **IP Filtering**

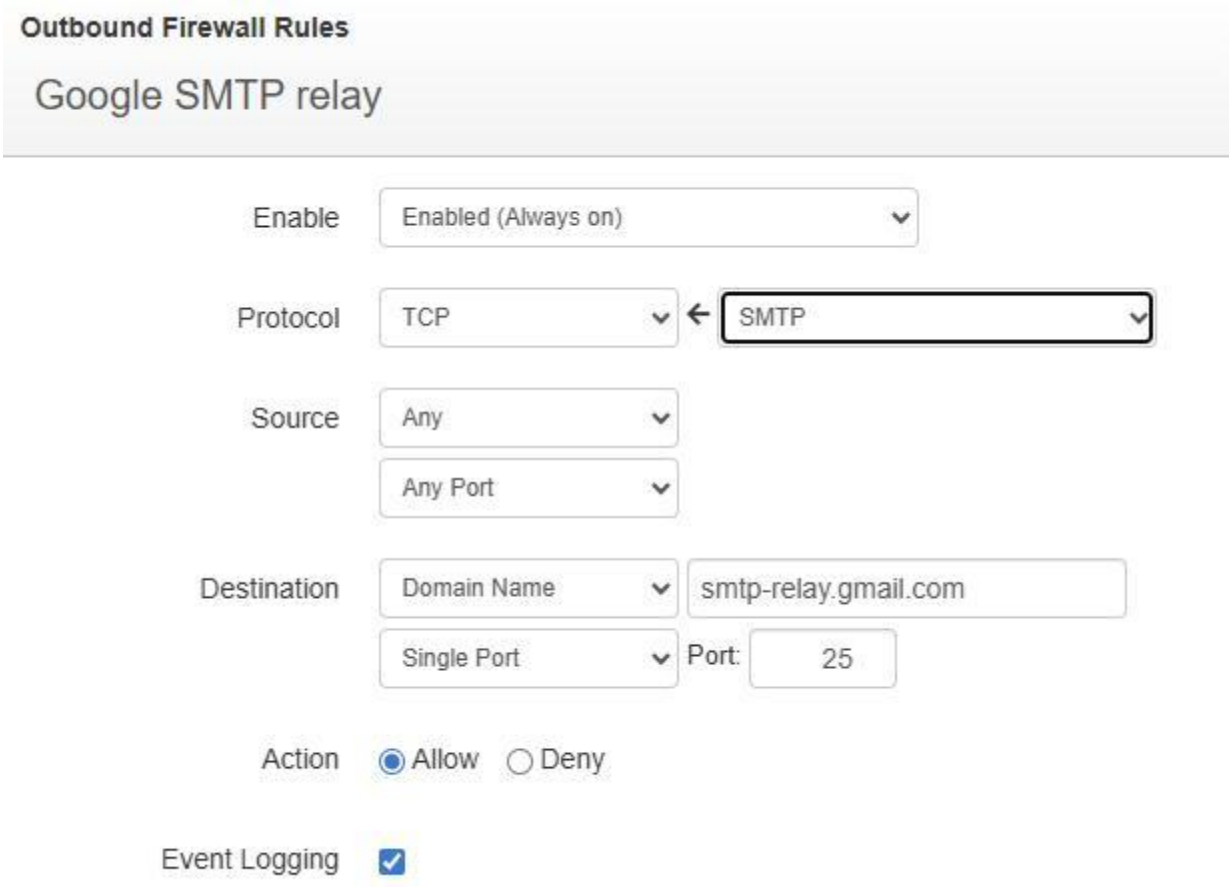
- **Port Filtering**



Step 3: Create a Blocking Rule

Enter the following settings:

Setting	Value
Rule Name	**Block SMTP**
Protocol	**TCP**
Direction	**Outgoing / LAN to WAN**
Start Port	**25**
End Port	**25**
Action	**Deny / Block**
Status	**Enabled**



Step 4: Verify the Rule

Check that your router's firewall or port-filtering page displays:

- **Rule: Block SMTP**
- **Protocol: TCP**
- **Port: 25**
- **Direction: Outgoing**
- **Action: Block**
- **Status: Enabled**

Blocking outgoing traffic on port 25 prevents devices in the network from directly sending emails through standard SMTP connections. This helps reduce spam emails sent by infected devices or malware operating inside the network.

4. **Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.**
   Hint: Focus on source/destination IPs and ports in the rule set.

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No.	Source IP	Destination IP	Port	Service
1	192.168.1.5	172.217.166.78	80	HTTP
2	10.0.0.10	34.201.23.1	22	SSH
3	192.168.1.7	8.8.8.8	53	DNS

Remember:

- **Port 80 = HTTP**
- **Port 22 = SSH**
- **Port 53 = DNS**

Without the firewall rules, we cannot correctly decide whether a packet is allowed or blocked.

5. **Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.**

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1. Blocking Unauthorized Traffic: Firewalls block unauthorized users and suspicious connections from accessing company servers and databases.

2. Preventing DDoS Attacks: Firewalls filter excessive and suspicious network requests to prevent servers from becoming overloaded or unavailable.

3. Blocking Malicious Requests: Packet filtering checks IP addresses and port numbers to block suspicious traffic and prevent unauthorized access.